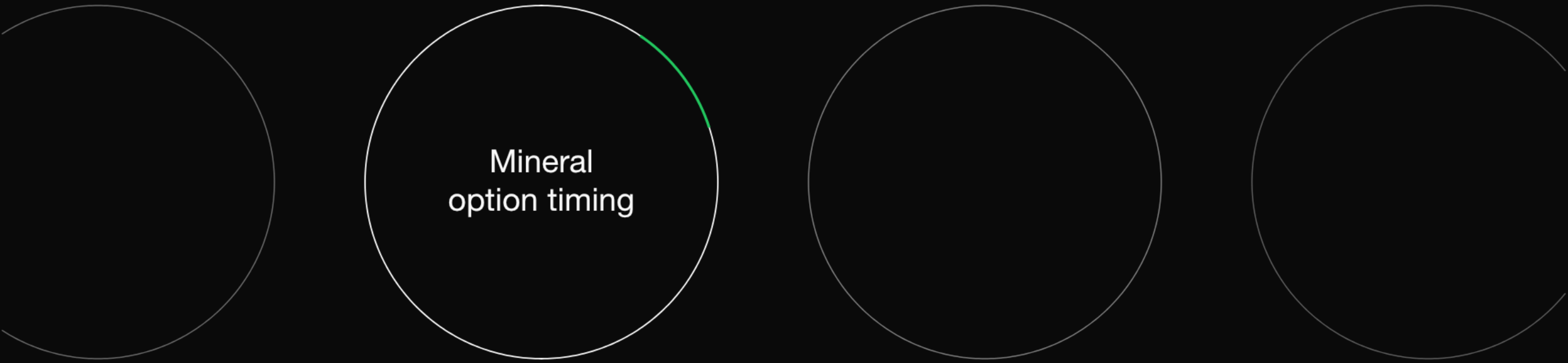
OXHack 2026
Mineral option timing

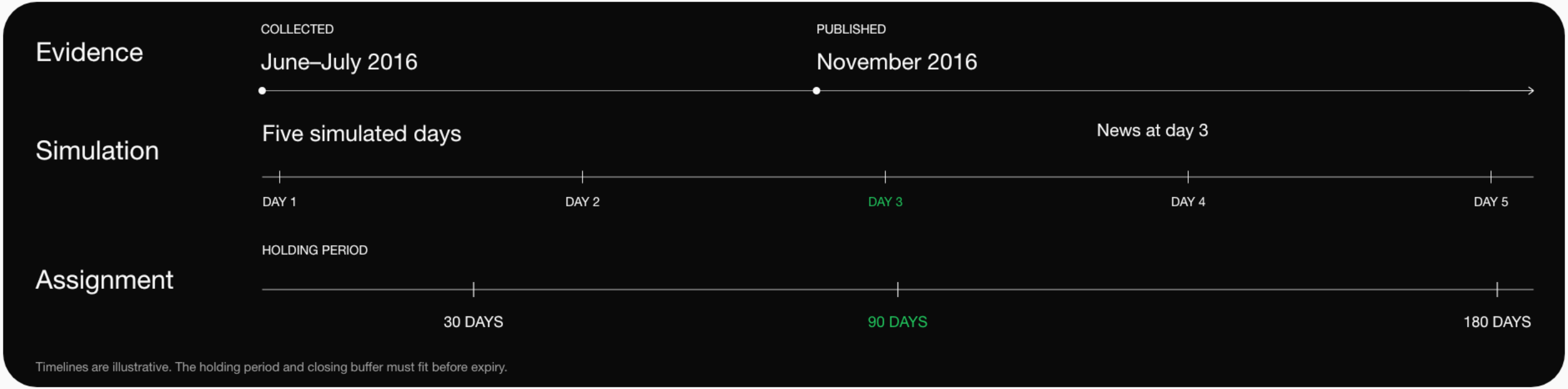


Local prototype

25 October 2026

Mineral evidence, changing conditions and time to exit belong in the same decision.

Our prototype investigates an assumed mineral option
and tests whether it can produce cash within the time available.



Timelines are illustrative. The holding period and closing buffer must fit before expiry.

We begin with

67

measured sediment samples from Tok River, Alaska.

A has 33 samples. B has 34.
Primary gold assays screen the evidence.

Recoverable ounces remain a separate assumption.

The evidence explorer

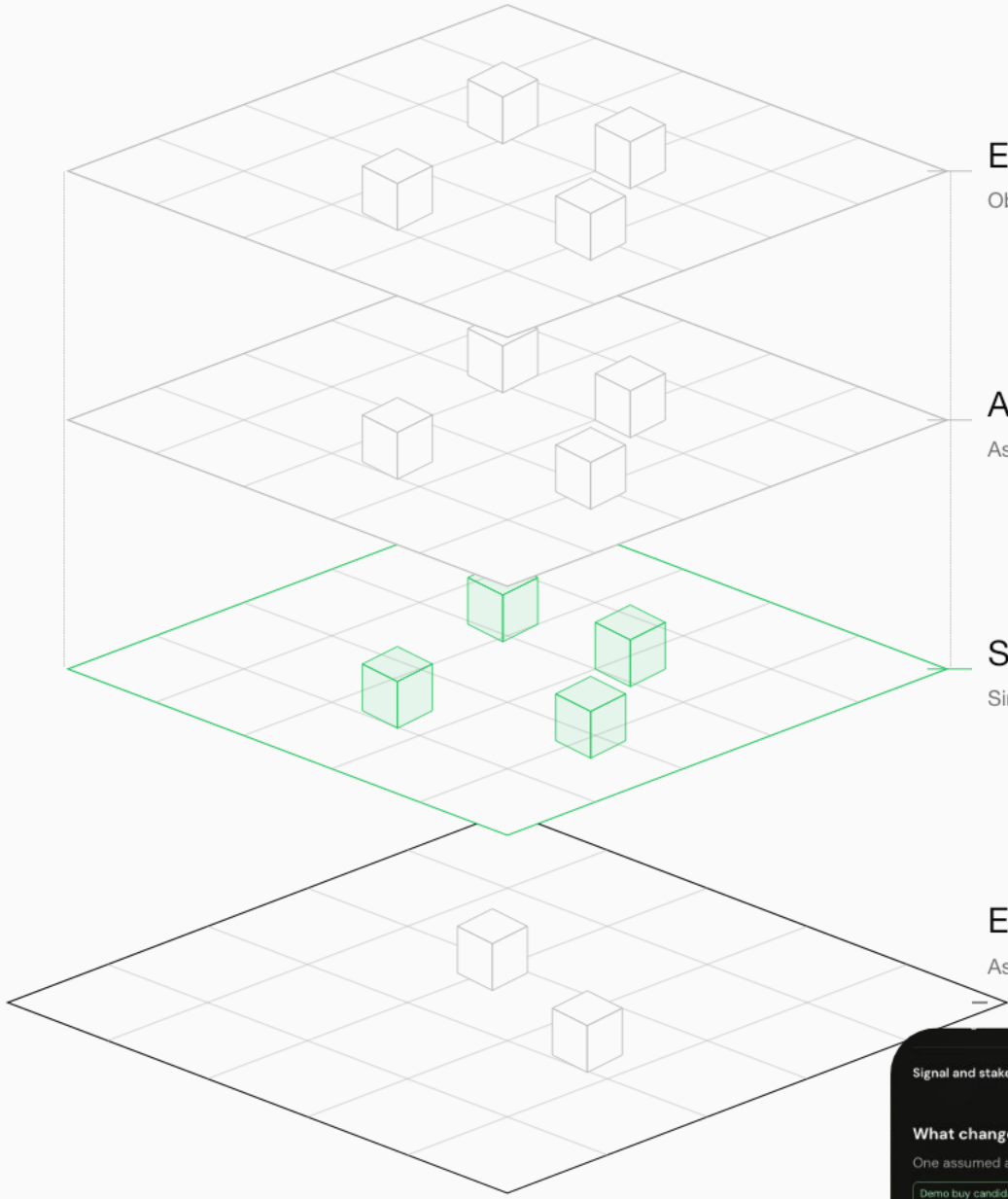


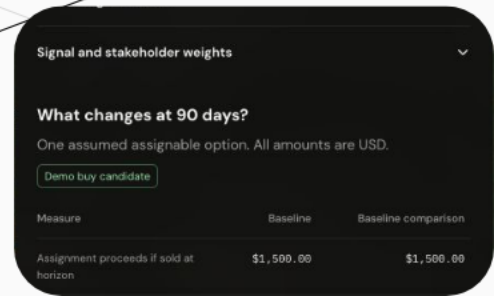
Alaska DGGS RDF 2016-10. Comparison areas, not available listings.



The algorithm connects mineral evidence and explicit assumptions to an assignment decision.

Observations screen evidence.
Simulations test reactions.
Economics remain assumed.





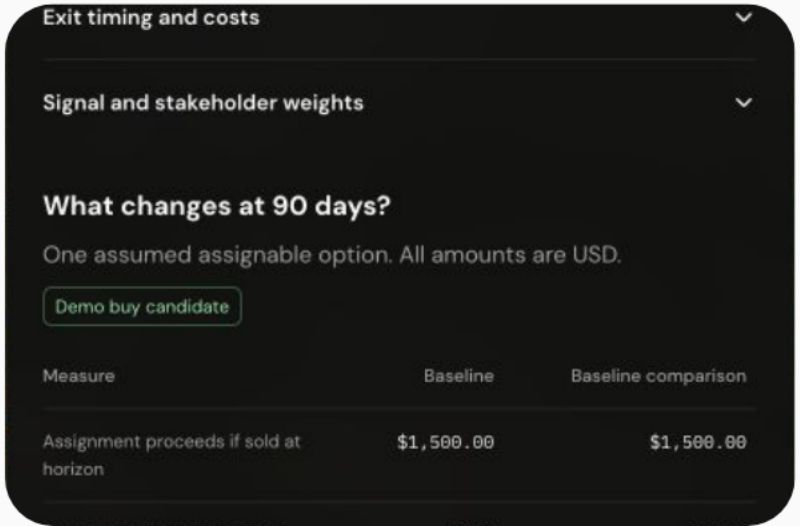
Recorded stakeholder reactions let us inspect how a news scenario changes the model.

Mine approval
–0.0252 support contrast

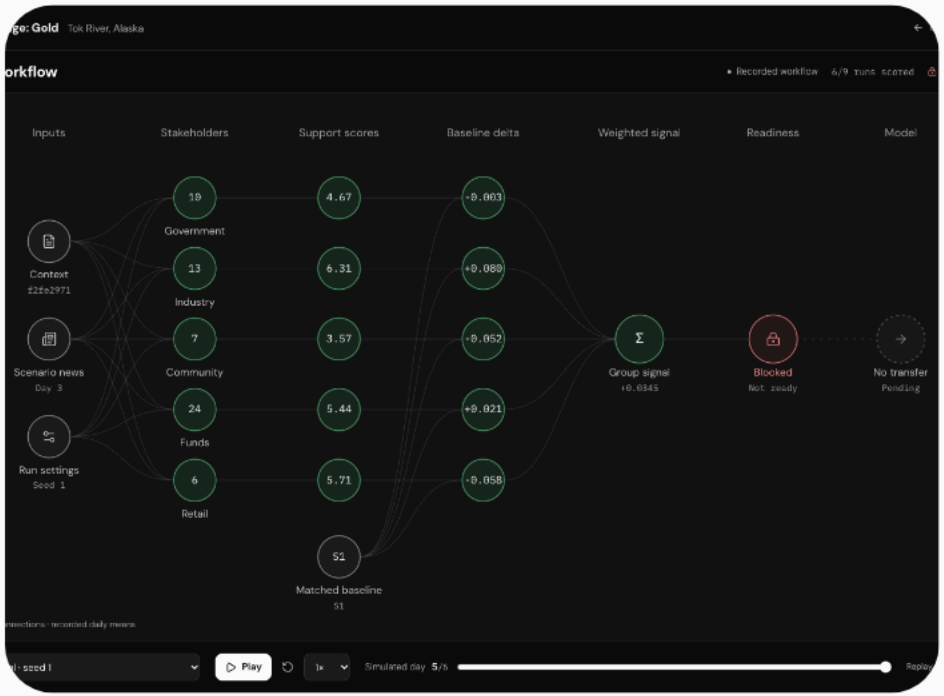
6 of 9 planned runs scored



Matching Claude sports-placebo reference, days 3–5.
Normalized support scale –1 to +1.



Court blockage
+0.0307 support contrast



Recorded research context. Checks failed.
Financial mapping is illustrative.

At 90 days, the same hypothetical option has

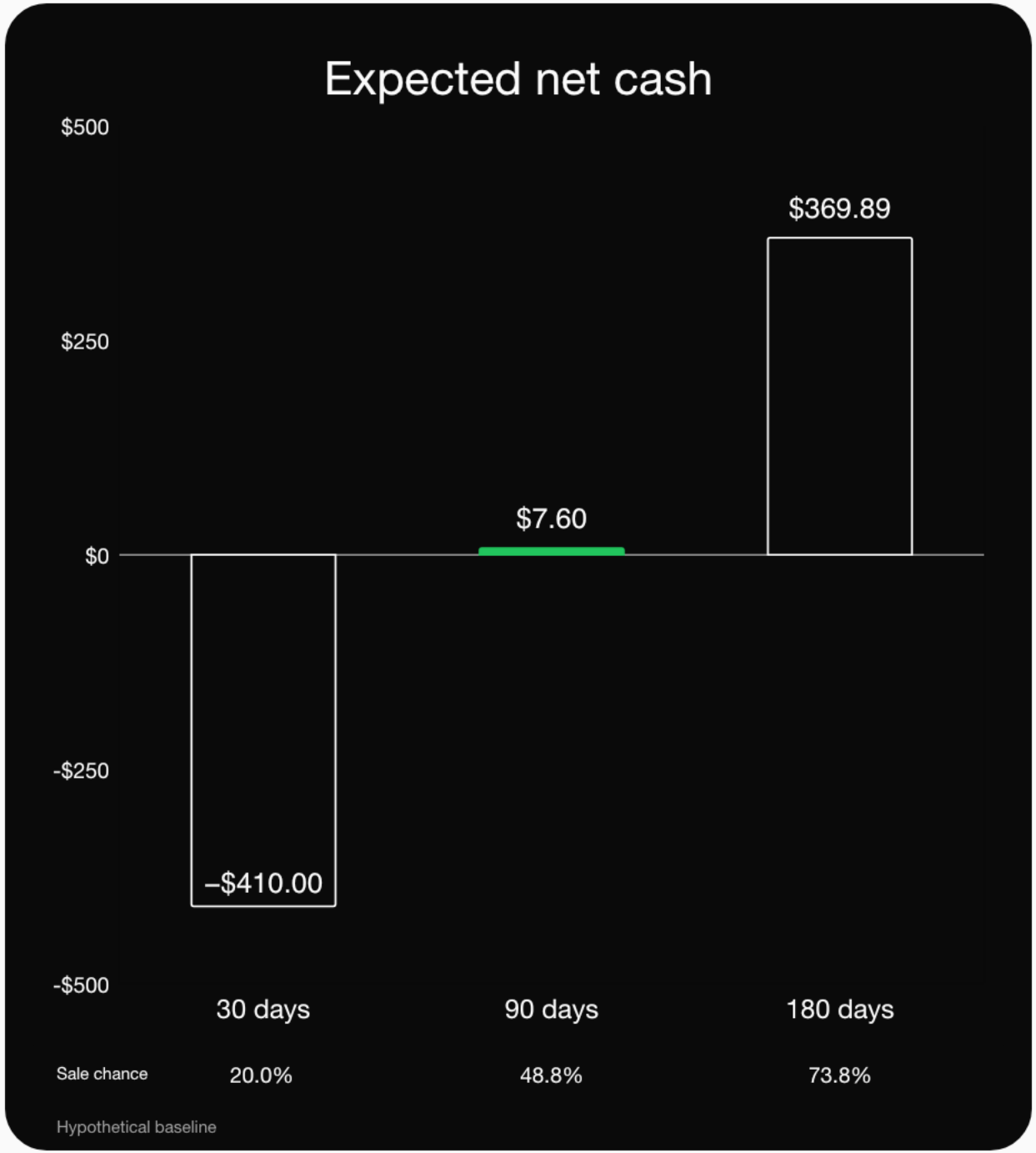
$7.60

expected net cash, while

51.2%

of paths have no sale.

$700 upfront. Cash if unsold: –$700.
20% assumed conditional sale chance per 30 days.
Fixed assignment price. All amounts USD.





Every decision can be traced to its evidence, scenario and timing assumptions.

Inspect the evidence. Compare a scenario. Change time. Explain the result.

Live demonstration

Financial outputs use hypothetical assumptions.

Imagery: Esri, Maxar, Earthstar Geographics. Terrain: Mapzen.

The 90-day cash calculation

Hypothetical cash branches after all entry and exit costs. All amounts USD.

Outcome by day 90	Probability	Net cash	Weighted cash
Sale	48.8%	+$750.00	+$366.00
No sale	51.2%	–$700.00	–$358.40
Expected net cash	**100.0%**		**+$7.60**

$$0.488 \times \$750 + 0.512 \times (-\$700) = \$7.60$$

Median cash: –$700. Break-even premium: $657.60.
A $660 premium changes expected cash to –$2.40 and the decision to skip.

Evidence, simulations and remaining validation

The prototype retains the provenance of each input.

Observed	Simulated	Assumed
67 measured sediment samples	Five stakeholder groups	Option terms and recoverable ounces
Checked land-source context	Six scored five-day runs	Sale chance per 30-day interval
Historical gold records	Research diagnostics failed	Capped financial scenario transfer

Next: chronological validation with dated option prices, completed assignments and unsold cases.

Predictive performance remains unmeasured.